Zinc doping of the Cuprates

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1 Introduction

J.C. Seamus Davis discussed with me the problem of zinc doping in the cuprates. It unknown why the superconductivity is killed by small doping. Seamus suggested I try modelling the situation by switching off the Hubbard U on the zinc site. The cuprate can be captured by a stacked SSH model, similar to that studied by Rosenberg and Manousakis [CITE: Rosenberg, Aryal PRB 100 104522 (2019)].

H. Alloul *et al* review Defects in correlated metals and superconductors [CITE:arXiv:0711.0877v1 (2018)].

2 Topology

Rosenberg and Manousakis [CITE: Rosenberg, Aryal PRB 100 104522 (2019)] showed topological modes when the interdimer hopping is different from the intradimer hopping, with equal chemical potential.

Have they checked with a rigid shift?

Any nonzero CDW-like density in the Fock channel would give rise to superconductivity.

3 Review of defects in correlated metals

H. Alloul *et al* [CITE:arXiv:0711.0877v1 (2018)]

4 Reproducing results

The normal state Hamiltonian is given by

$$\begin{aligned}\hat{\mathcal{H}}_0 = &\sum_{0\leq x,y\leq N}\sum_{\sigma}\sum_{m\in\{A,B\}}\left[-\mu_m \hat{c}^{\sigma,m\dagger}_{x,y}\hat{c}^{\sigma,m}_{x,y}\right] \\ &- t\sum_{0\leq x,y\leq N}\sum_{\sigma}\left[\hat{c}^{\sigma,A\dagger}_{x,y}\hat{c}^{\sigma,B}_{x,y} + \hat{c}^{\sigma,A\dagger}_{x,y}\hat{c}^{\sigma,B}_{x,y+1} + \hat{c}^{\sigma,B\dagger}_{x,y}\hat{c}^{\sigma,A}_{x+1,y+1} + \hat{c}^{\sigma,B\dagger}_{x,y}\hat{c}^{\sigma,A}_{x+1,y} + \text{h.c.}\right] \quad (1)\end{aligned}$$

where the fermion operator $\hat{c}^{\sigma,m\dagger}_{x,y}$ creates an operator on atom $m \in \{A, B\}$ within the unit cell located at $\mathbf{R}_{x,y} = x\mathbf{b}_0\hat{x} + y\mathbf{b}_1\hat{y}$, where $\mathbf{b}_0, \mathbf{b}_1$ are the lattice basis vectors, $\hat{n}^{(\alpha)\dagger}_{x,y}$ is the number operator, μ is the chemical potential and s is a rigid shift of the atomic potentials. The intra- and inter-unit-cell hoppings strengths in the $\hat{x}$ direction are given by v and w, which form the terms of the standard SSH model. These 1D SSH chains are coupled by a diagonal hopping t_d, which form a two-dimensional stacking. The lattice is depicted with $N = 3$ unit cells in Figure 1. Following [CITE: Rosenberg and Manousakis arXiv:2203.12004v1], without loss of generality, we set $v = 0.6, w = 1.2$, the topological phase of the SSH model, and $t_d = 0.9$.

The solution to the Hamiltonian is given by the Green's function, which we remind the reader, is given by

$$\hat{\mathcal{G}}(\omega - i\epsilon, x, y) \equiv \sum_n \left[\frac{\psi_n(x,y)\psi_n(x,y)^\dagger}{\omega - i\epsilon - E_n}\right], \quad (2)$$

where $\psi_n(x, y)$ is the eigenvector associated with the nth eigenenergy, resolved at the point on the lattice (x, y). The Green's function matrix has a component for each atom. The Green's function is experimental observables through the *local density of states*,

$$\text{LDOS}(\omega; x, y) \equiv -\frac{1}{\pi}\Im\hat{\mathcal{G}}(\omega - i\epsilon, x, y), \quad (3)$$

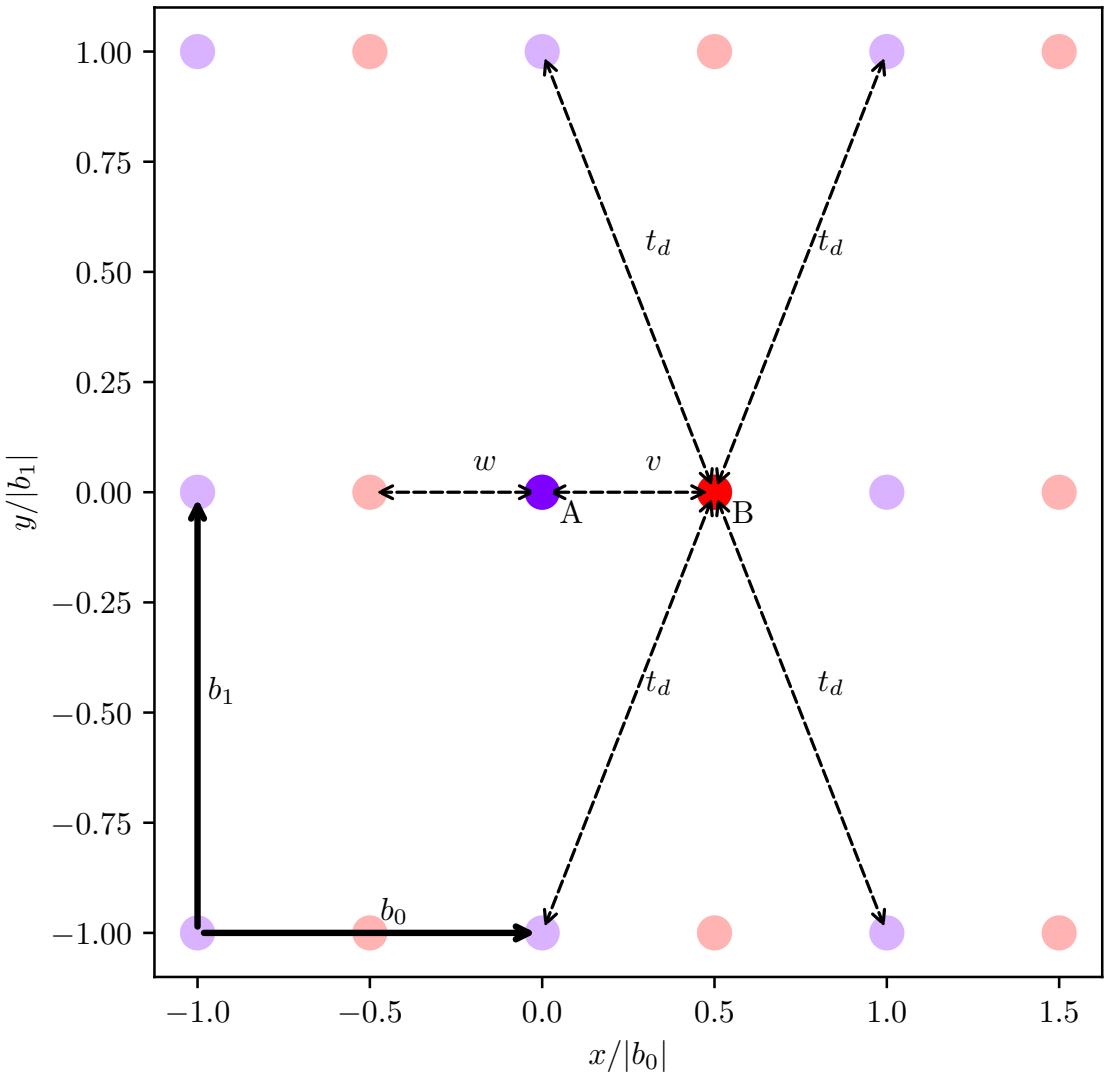

Figure 1: Two-dimensional multiorbital Weyl SSH model. Along the $\hat{x}$-direction, atomic sites A and B within the unit cell form an SSH chain, with intracell hopping v and intercell hopping w. An additional interchain hopping parameter couples the chains into a two-dimensional stack. Following Rosenberg and Manousakis, we study a diagonal hopping t_d, which is a Type II extension of the SSH model in the classification according to Bo-Hung [CITE: Two-Dimensional Extended Su-Schrieffer-Heeger Model; Masters Thesis; Chen, Bo-Hung; July 2018]. The basis for the lattice is given by the vectors b_0 and b_1.

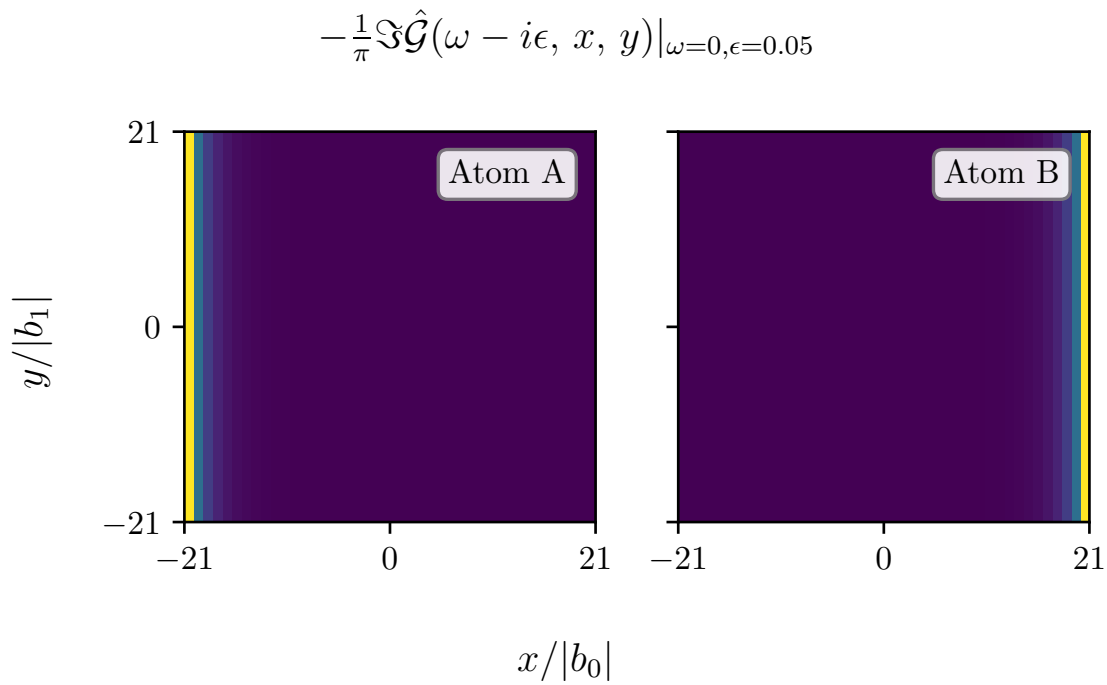

Figure 2: The local density of states of a Weyl SSH model, that is, a spinless square lattice with 43×43 lattice cells, with chemical potential $\mu/t = 0.00$, zero shift $s = 0$ and applied bias $\omega = 0$ and spectral resolution $\epsilon = 0.05$. The SSH chains are in their topological phase $w = 1.2 > v = 0.6$, with diagonal hopping $t_d = 0.9$, which does gives rise to no topological modes perpendicular to the chains. Notice the zero bias density at the left edge for atom A and on the right for atom B, a signature of topological modes.

which is a measure of the density of states at a given point, at a specified applied bias energy ω and resolution ϵ. The local density of states for each atomic site with the unit cells are given in Figure 2, showing the existence of zero energy edge states.

Resolving the local density states at the level of the unit cell (summing over the atoms) and integrating over the y-axis, we study the *spectrum* of the system. The spectrum shows edge modes present at zero energy at the edges, which are not present in the bulk. These are distinct signatures of Majorana modes. In addition to localised density at the edge, there exists a 'tail' of density which the authors investigate further.

Majorana modes exist pinned to *Dirac points*, which are the cross point of two bands. Studying the band structure in Figure 11, we observe a zero-mode between the Dirac points, which is known as a *Bogoliubov-Fermi arc*.

In order to determine whether the modes are topological modes, it must be shown

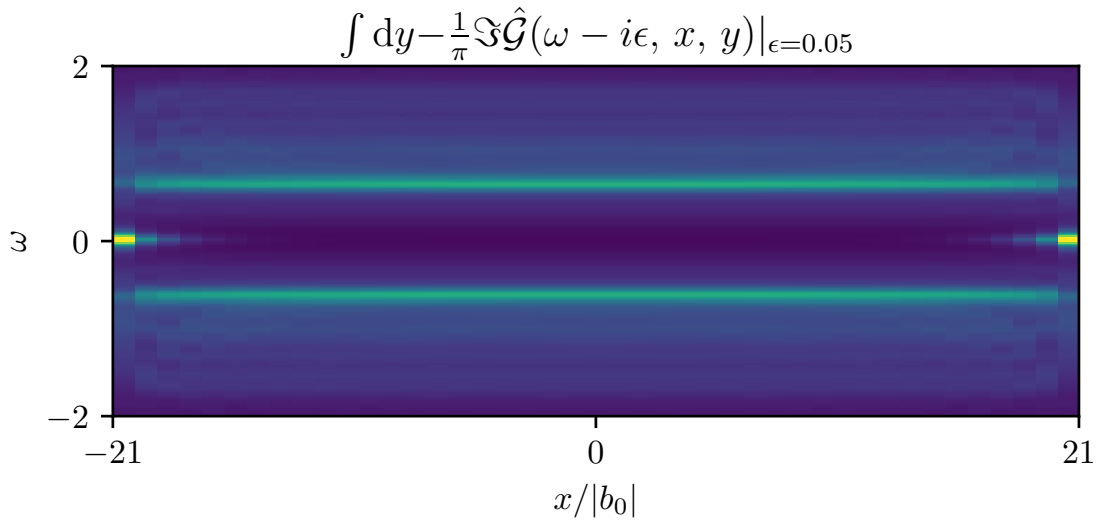

Figure 3: The spectrum of the Weyl-SSH model, in its normal state, integrated over the direction perpendicular to the SSH chains and summed over the atomic sites within the lattice cell. Zero-energy modes are observed at the edges, with a faint tail of density.

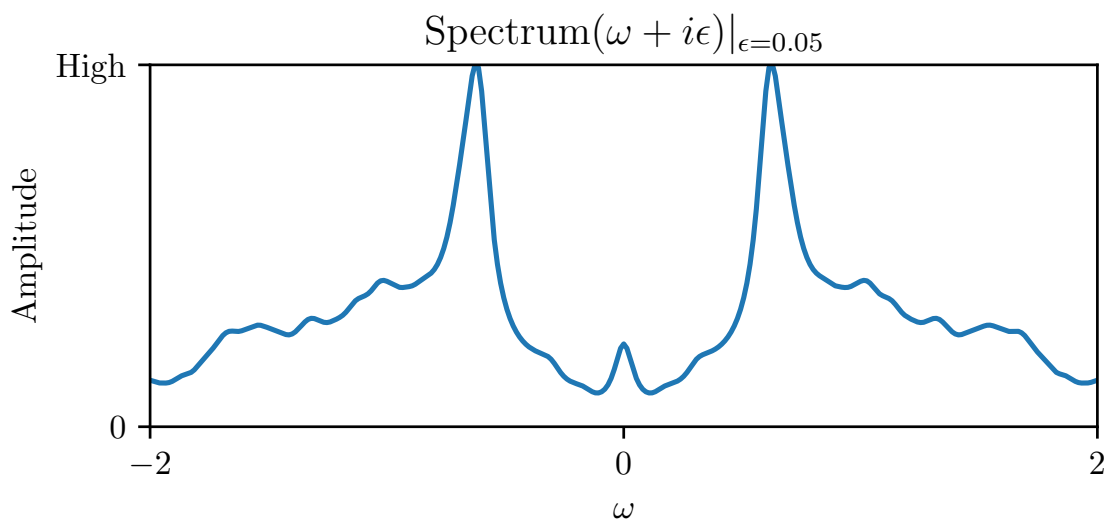

Figure 4: The spectrum of the Weyl-SSH model in the normal state with open-boundary conditions, summed over the atomic sites. The parameters are $\mu = 0.0, v = 0.6, w = 1.2$ and $t_d = 0.9$. We observe the topological mode as a density peak within the band gap.

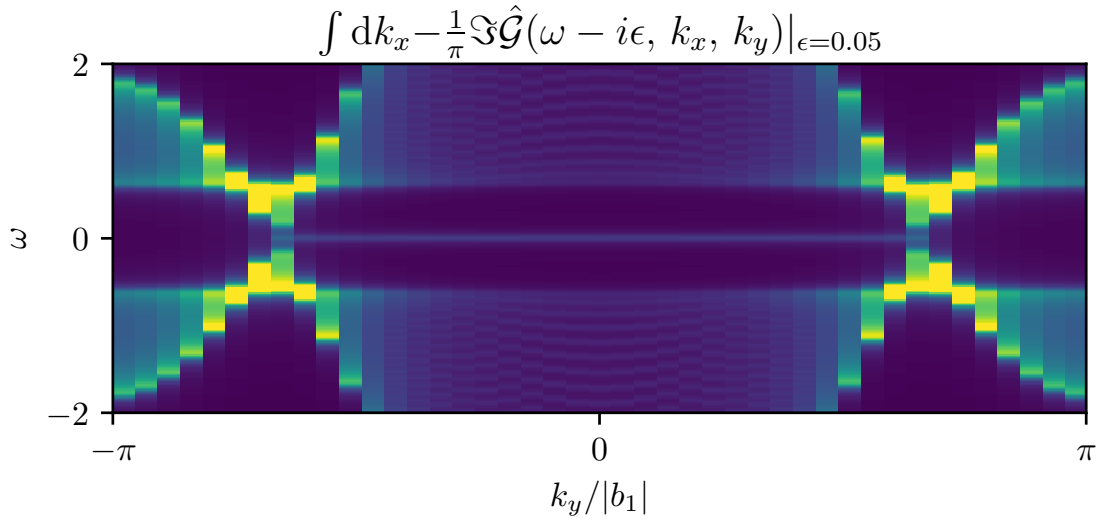

Figure 5: The spectrum of the Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. Dirac nodes are observed, which pin Majorana zero-modes. Between the Dirac nodes is a line of zero-energy states, known as a Bogoliubov-Fermi arc.

that they their wavefunctions are distributed nonlocally across the system. We plot the wavefunction via the related *spectral density* of the Green's function at the point k_y at which the Bogoliubov-Fermi arc and Majorana modes are are located, Figure 9. As a function of x-position, we observe that indeed, the modes are spread non-locally and that the arc tails.

We note that other hopping configurations are possible, leading either to topological modes along one pair of parallel edges, or along all edges; see [Two-Dimensional Extended Su-Schrieffer-Heeger Model; Bo Hung; Masters Thesis; National Taiwan Normal University] .

5 Interactions and mean-field solution

We add an interaction term to model multiorbital superconductivity within the unit-cell, U_v, and an interorbital intercell Coulomb repulsion term, U_w. The interacting contribution to the Hamiltonian is

$$\hat{\mathcal{H}}_I = -U_v \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x,y} \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x,y} + U_w \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x-1,y} \hat{c}^{(B)}_{x-1,y} \hat{c}^{(A)}_{x,y}. \tag{4}$$

The mean-field Hamiltonian is

$$\hat{\mathcal{H}}'_I = \sum_{x,y} \Bigg[\sum_m \Big[\phi_m \hat{n}^{(m)}_{x,y} \Big] + \Phi_v \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x,y} + \Phi_w \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x-1,y} + \Delta_v \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x,y} + \Delta_w \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x-1,y} + \Delta^*_v \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x,y} + \Delta^*_w \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x-1,y} \Bigg], \tag{5}$$

where ϕ_v is the *Hartree* term which renormalises the chemical potential, allowing for a Peirl's instability. Φ_v is a *Fock* term, describing renormalistion of the hopping parameter v. The additional *Gorkov* terms Δ_v capture fermion-fermion pairing, relevant to theories of magnetism and superconductivity. These terms are known as the method of *(superconducting) density functional theory, SC-DFT*. The self-consistent solution to the mean field is the saddle point $\langle \hat{\mathcal{H}}_I \rangle = \langle \hat{\mathcal{H}}'_I \rangle$. The solutions are

$$\rho_m = (-U_v + U_w)) \langle \hat{n}^{(m')}_{x,y} \rangle, \tag{6}$$

$$\Phi_v = -U_v \langle \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)}_{x,y} \rangle, \tag{7}$$

$$\Phi_w = U_w \langle \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)}_{x-1,y} \rangle, \tag{8}$$

$$\Delta_v = +U_v \langle \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x,y} \rangle, \tag{9}$$

$$\Delta_w = -U_w \langle \hat{c}^{(A)}_{x,y} \hat{c}^{(B)}_{x-1,y} \rangle, \tag{10}$$

$$\Delta^*_v = +U_v \langle \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x,y} \rangle, \tag{11}$$

$$\Delta^*_w = -U_w \langle \hat{c}^{(A)\dagger}_{x,y} \hat{c}^{(B)\dagger}_{x-1,y} \rangle, \tag{12}$$

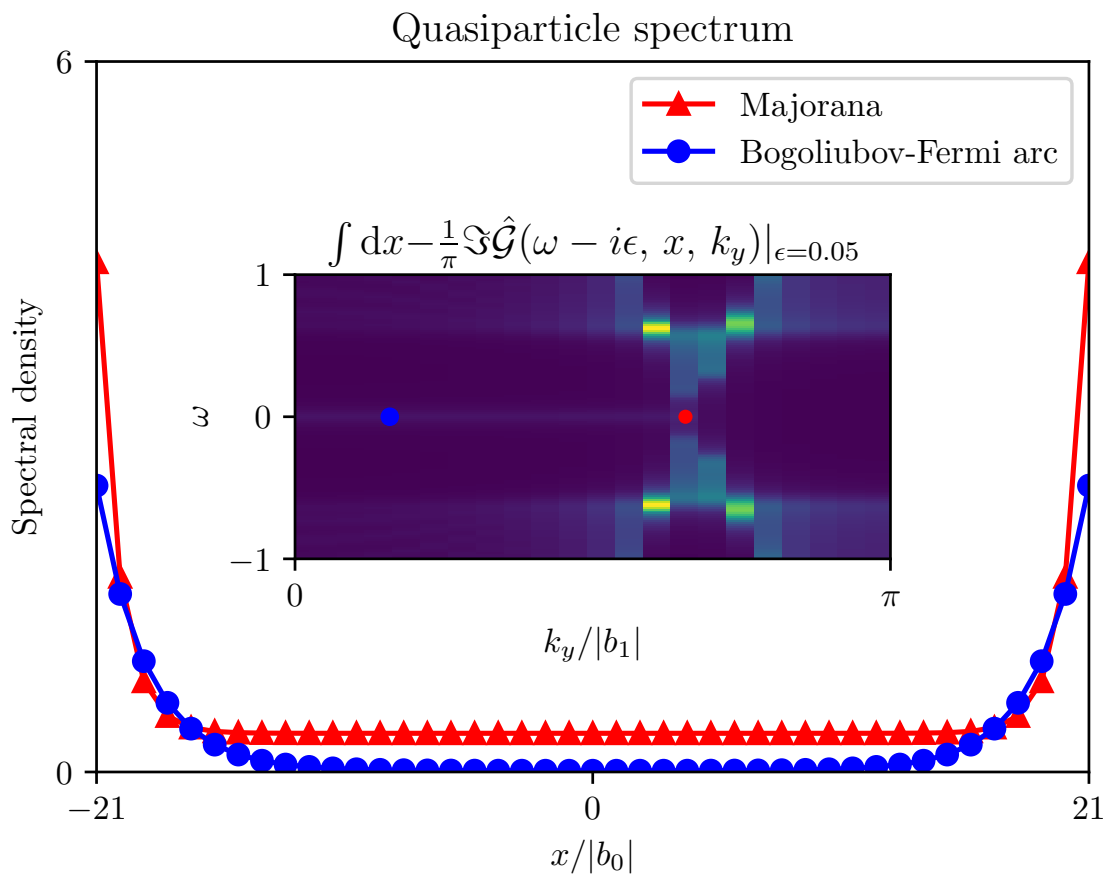

Figure 6: The spectrum of the Majorana and Bogoliubov-Fermi arc quasiparticle modes, at zero-applied bias, with the vertical momentum specified by the (colour-coordinated) spectral plot. The quasiparticle density is plotted over the direction parallel to the SSH chains. The Majorana mode (red) is localised at the edges. The Bogoliubov-Fermi arc, in contrast, tails.

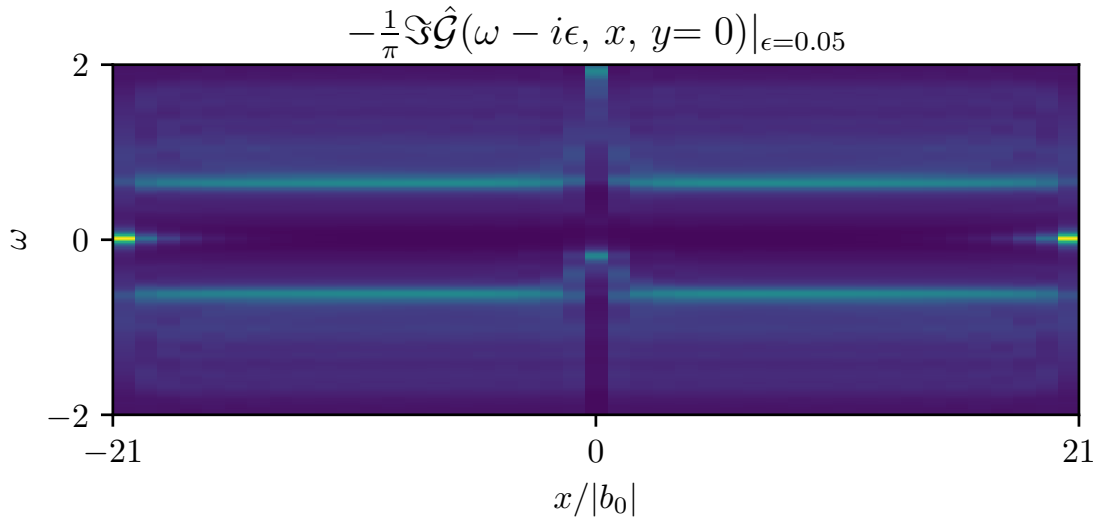

Figure 7: The local density of states along a line parallel to the axis of the dimers through a central impurity. The chemical potential is $\mu/t = 0.00$ and the lattice is 43×43 with periodic boundary conditions along the axis perpendicular to the dimers, and open boundary conditions along the axis. The impurity has coupling strength $V/t = 1.21$. No additional zero-mode is seen.

where m' refers to the opposite atom to m. Note the sign of the Gorkov terms due Fermionic anticommutation.

For simplicity and because we are in particular interested in the multiorbital pairing due to U_v, we neglect the intercell interaction U_w.

6 Results with impurities

The impurity is located at the centre with coupling strength $V = 1.21$.

The x-axis is glued to periodic boundary conditions.

Our investigation shows no interesting interplay between a single impurity and topological modes, however, impurities can be used to generate quasiparticle-interference patterns as an experimental diagnosis of topological modes. Moreover, a wall of impurities

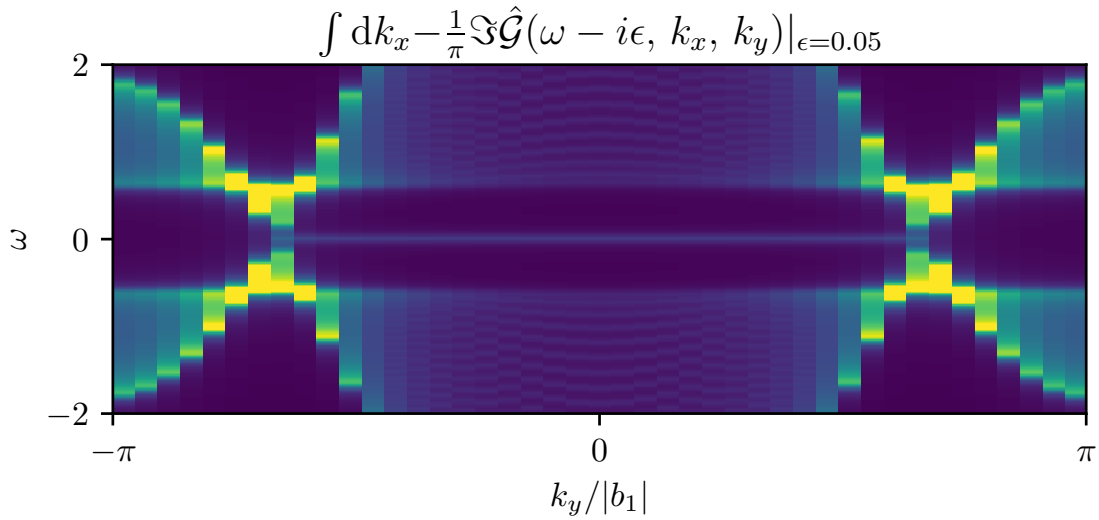

Figure 8: Band structure integrated over the axis of the SSH dimers, k_x, with open boundary conditions along the axis perpendicular to the dimers. No qualitative change in the spectrum is observed from the impurity-free model.

7 Quasiparticle-interference patterns for Majorana and Bogoliubov-Fermi arcs

8 Phase diagram

9 Wall impurities

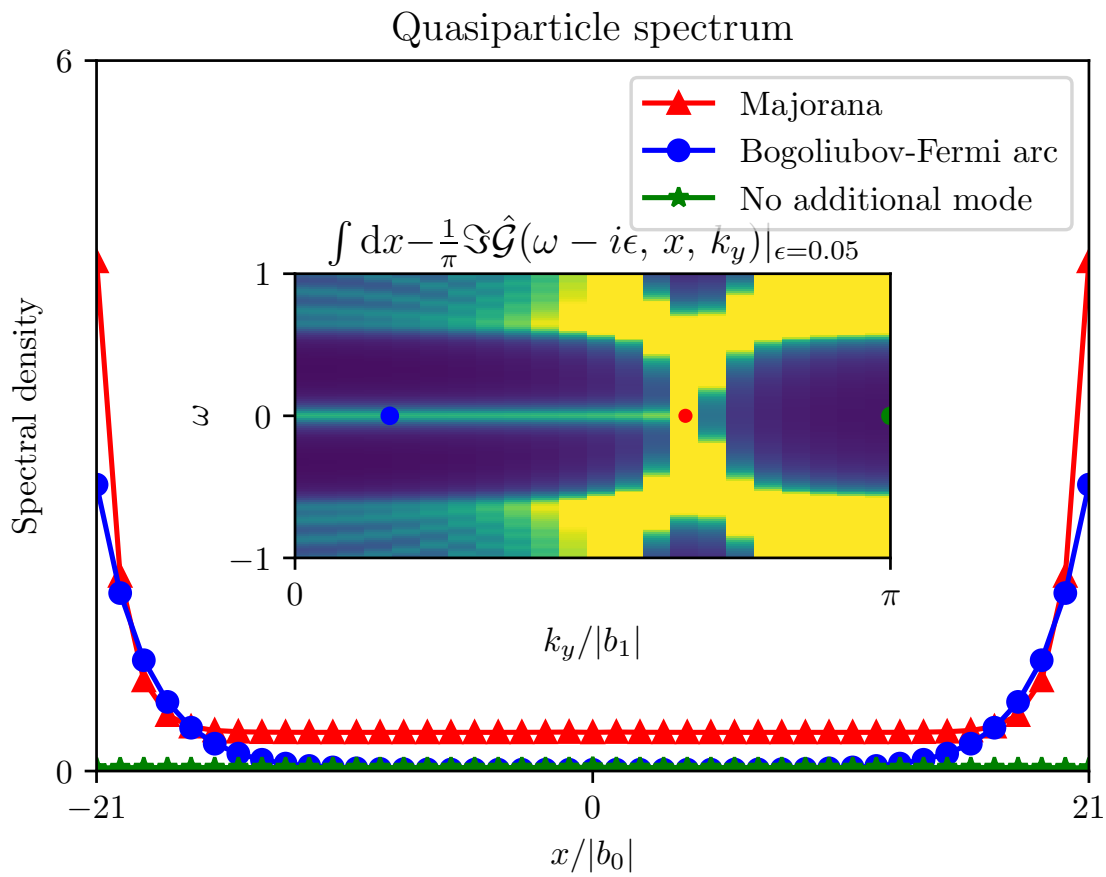

Figure 9: Topological modes in the presence of a single impurity. No localisation of the quasiparticles around the impurity are seen. Compare with Figure 12 –we observe an additional Bogoliubov-Fermi arc in a model with a wall of impurities. Note that the axis perpendicular to the dimers has been integrated over in the spectral density plot.

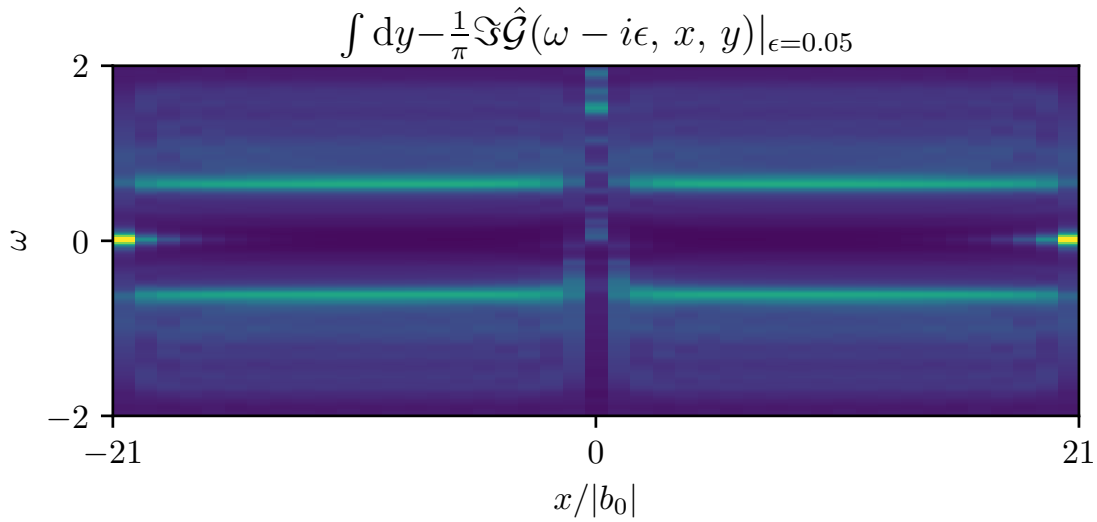

Figure 10: Local density of states with an impurity at the centre. The chemical potential is $\mu/t = 0.00$ and the lattice is 43×43 with periodic boundary conditions perpendicular to the axis of the dimers, and open boundary conditions along the axis parallel. The impurity wall runs perpendicular to the axis and through the middle, with coupling strength $V/t = 1.21$. Additional modes are seen although at such weak coupling, it is not clear whether they are gapped.

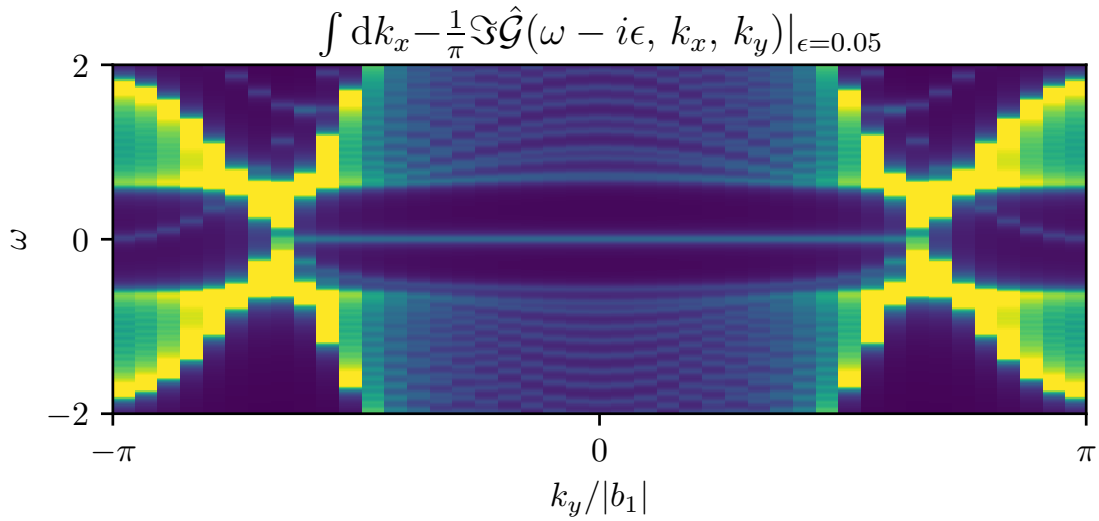

Figure 11: Local density of states integrated over the axis of the SSH dimers and summed over the intracell atomic sites show clearly that there exists an additional gapped mode due to the impurity wall. The chemical potential is $\mu/t = 0.00$ and the lattice is 43×43 with periodic boundary conditions along the axis perpendicular to the dimers, and open boundary conditions parallel. The impurity wall has coupling strength $V/t = 1.21$.

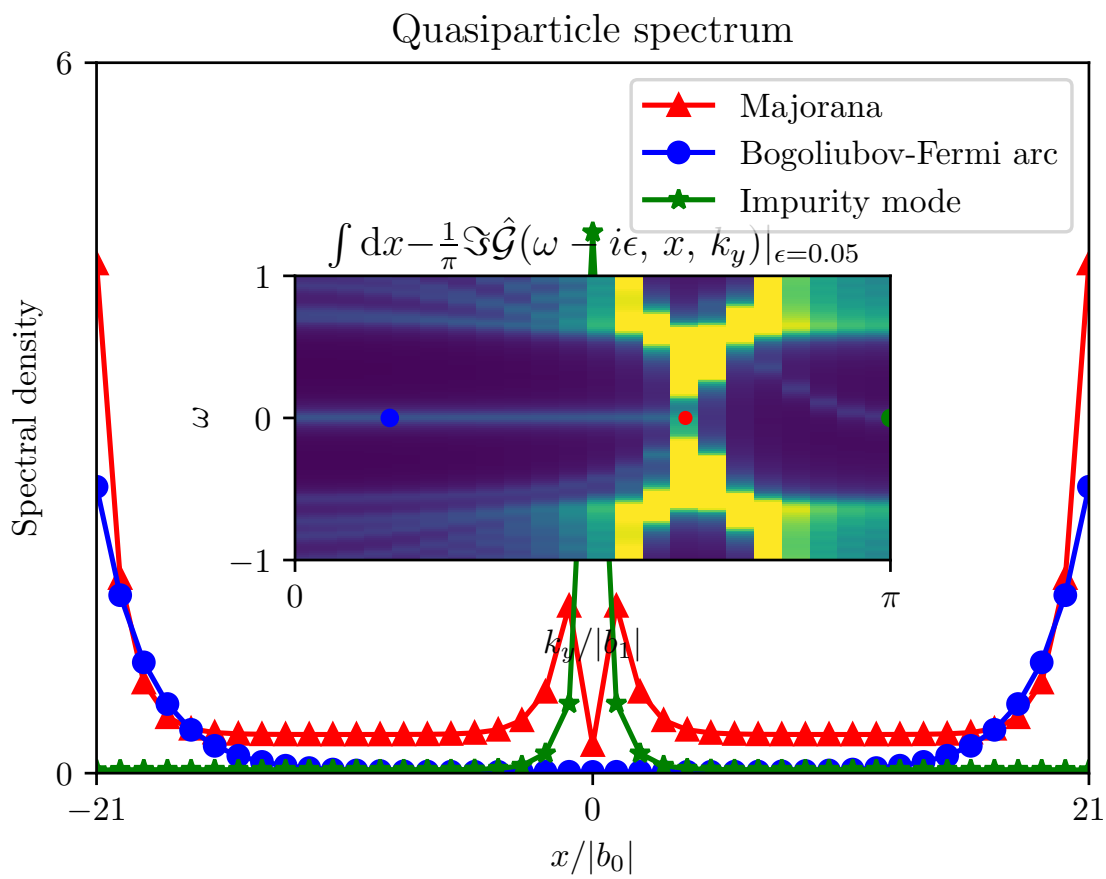

Figure 12: The majorana quasiparticle exhibits some pinning around the impurity wall and an additional arc mode exists at $k_y = \pi$, whose nature is to be investigated.

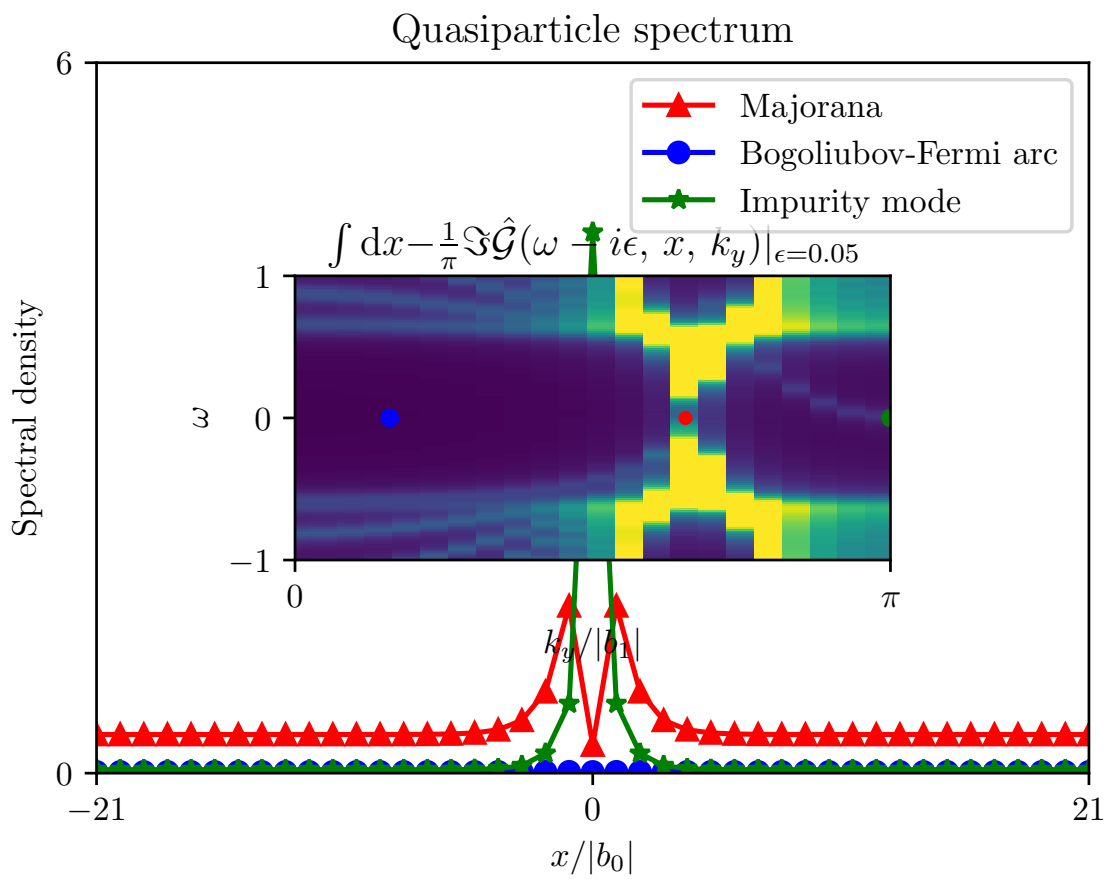

Figure 13: Investigation of the modes with periodic boundary conditions along both axes. The additional mode is shown to be due to the presence of a wall of impurities.

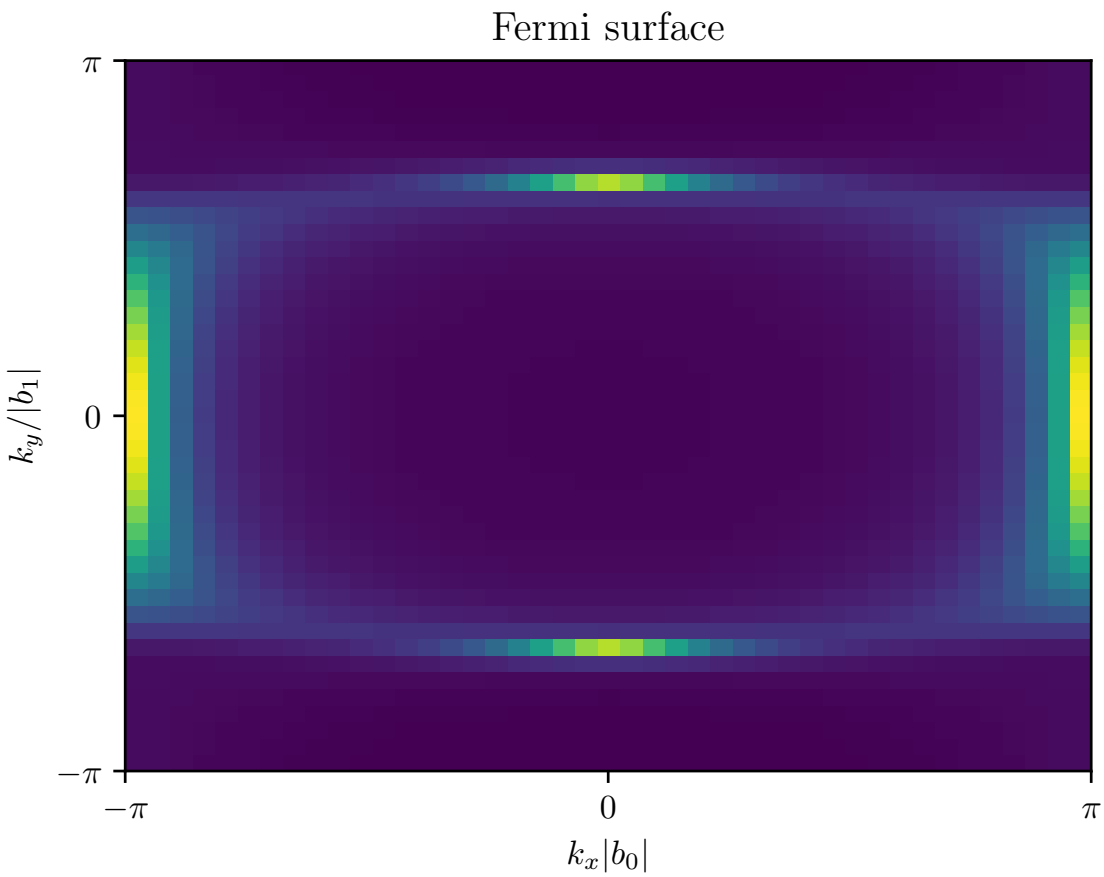

Figure 14: Fermi surface with a central impurity with coupling strength $V = 1.21$ and chemical potential $\mu = 0.0$, in the normal state. Notice density concentrated near $k_y \approx \pi/2$ corresponding to the Majorana mode, and density around $k_x \approx \pm\pi$ corresponding to the Fermi arc.

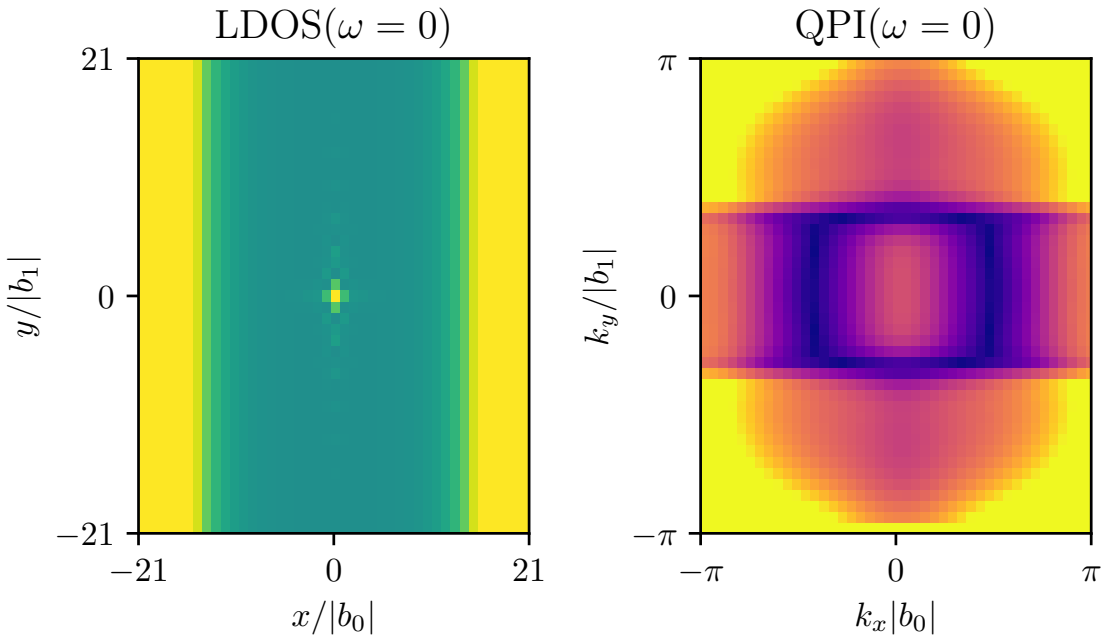

Figure 15: The local density of states and quasiparticle interference pattern for an 43×43 cell Weyl-SSH model with periodic boundary conditions along the vertical, and open boundary conditions along the horizontal, with a central impurity with coupling strength $V/t = 1.21$. The chemical potential is $\mu/t = 0.00$.

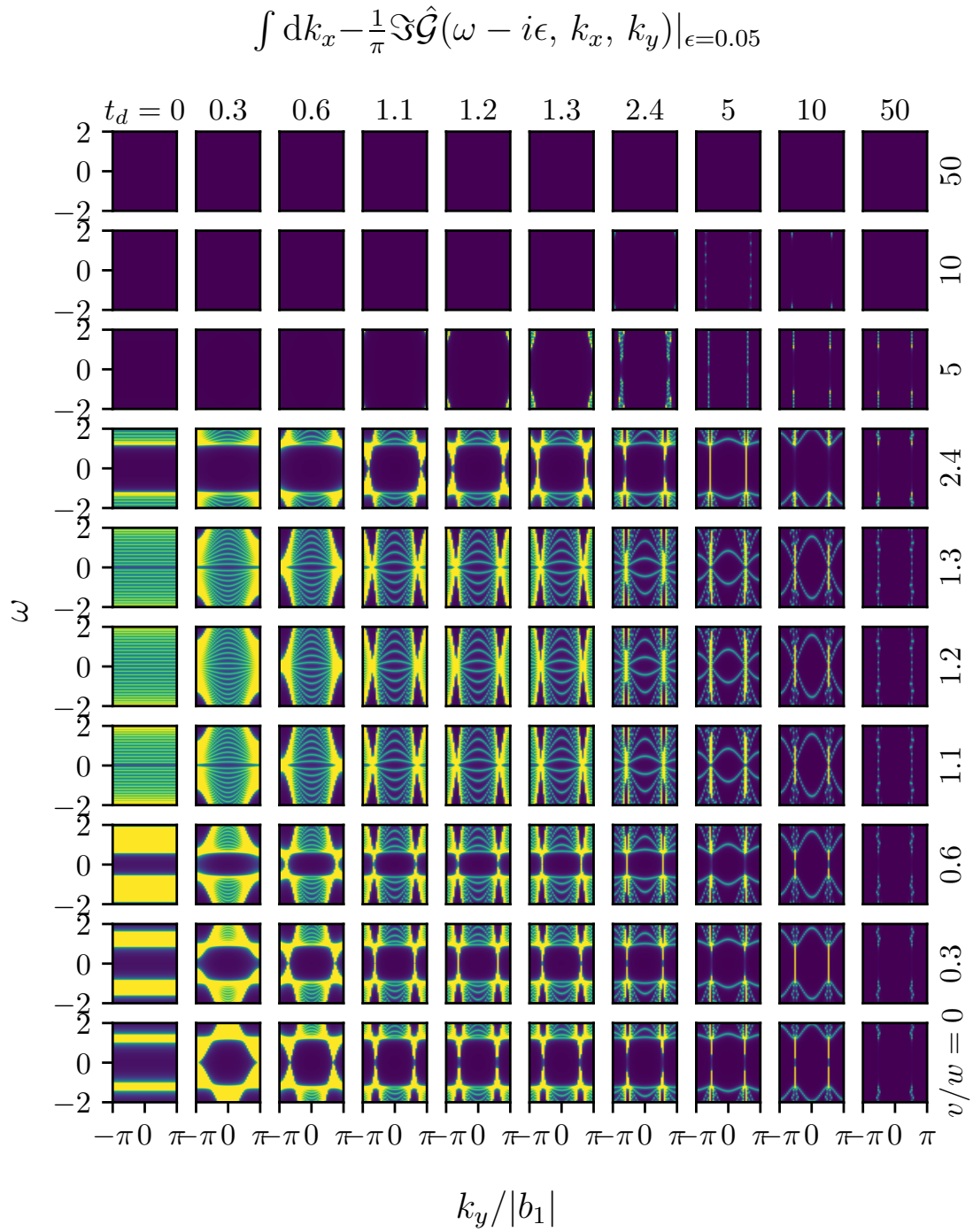

Figure 16: The spectrum of the simple Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. The model consists of an 43×43 lattice, with closed boundary conditions. The chemical potential is $\mu = 0.0$, the interdimer hopping is fixed $w = 1.2$, and the intradimer v and interchain t_d hoppings are varied. Compare with the model with open boundary conditions Fig **??**, and notice the universality when comparing to the NiC_2 model Fig **??**, **??**.

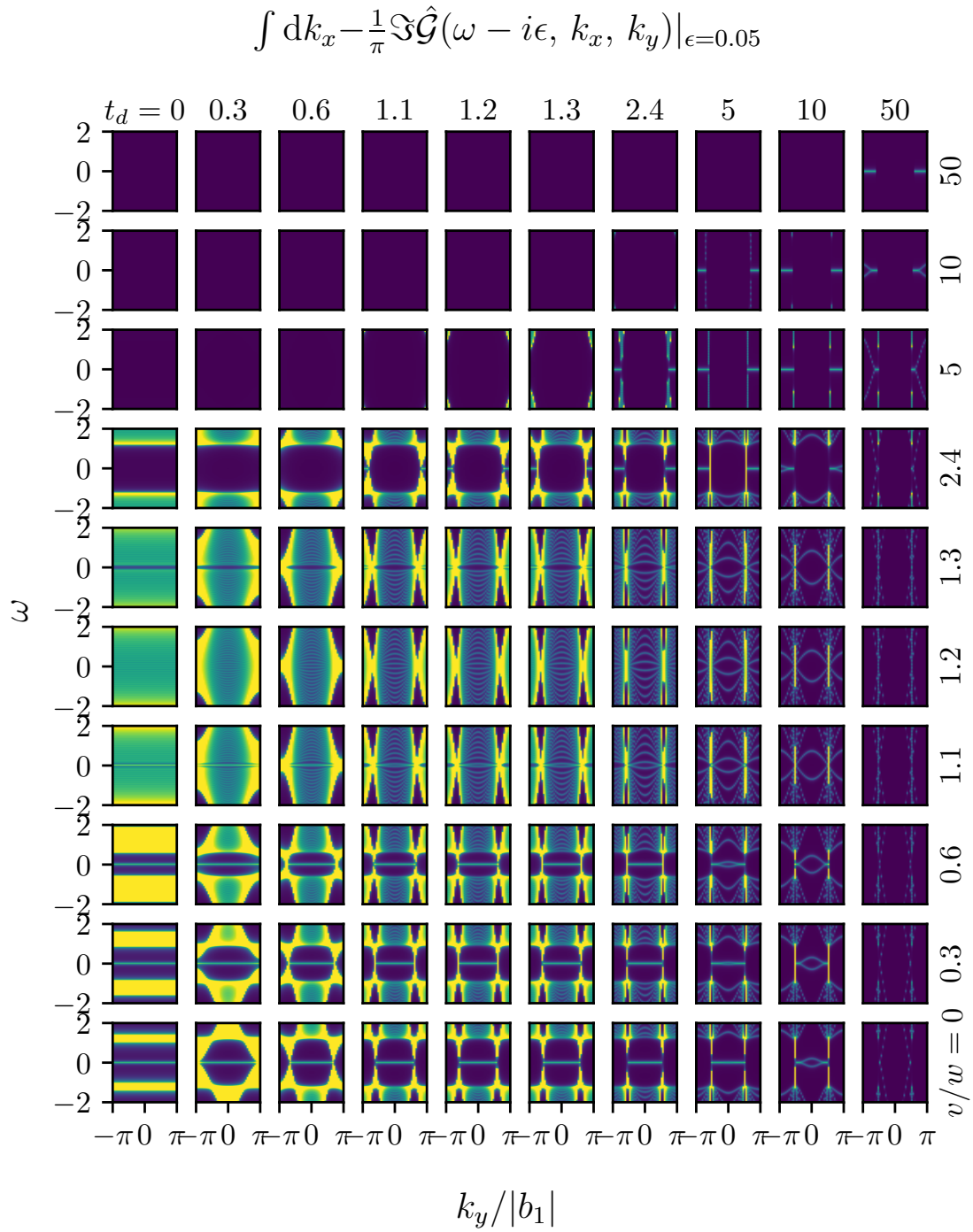

Figure 17: The model and quantities depicted as in Fig 16, but with the boundary conditions open. We observe the diagonal hopping t_d gives rise to two Dirac cones. In the SSH topological phase, a Bogliubov-Fermi arc connects them through the centre of the Brillouin zone. Then, in the SSH trivial phase, we find the Fermi arc instead passes through the edge of the Brillouin zone.

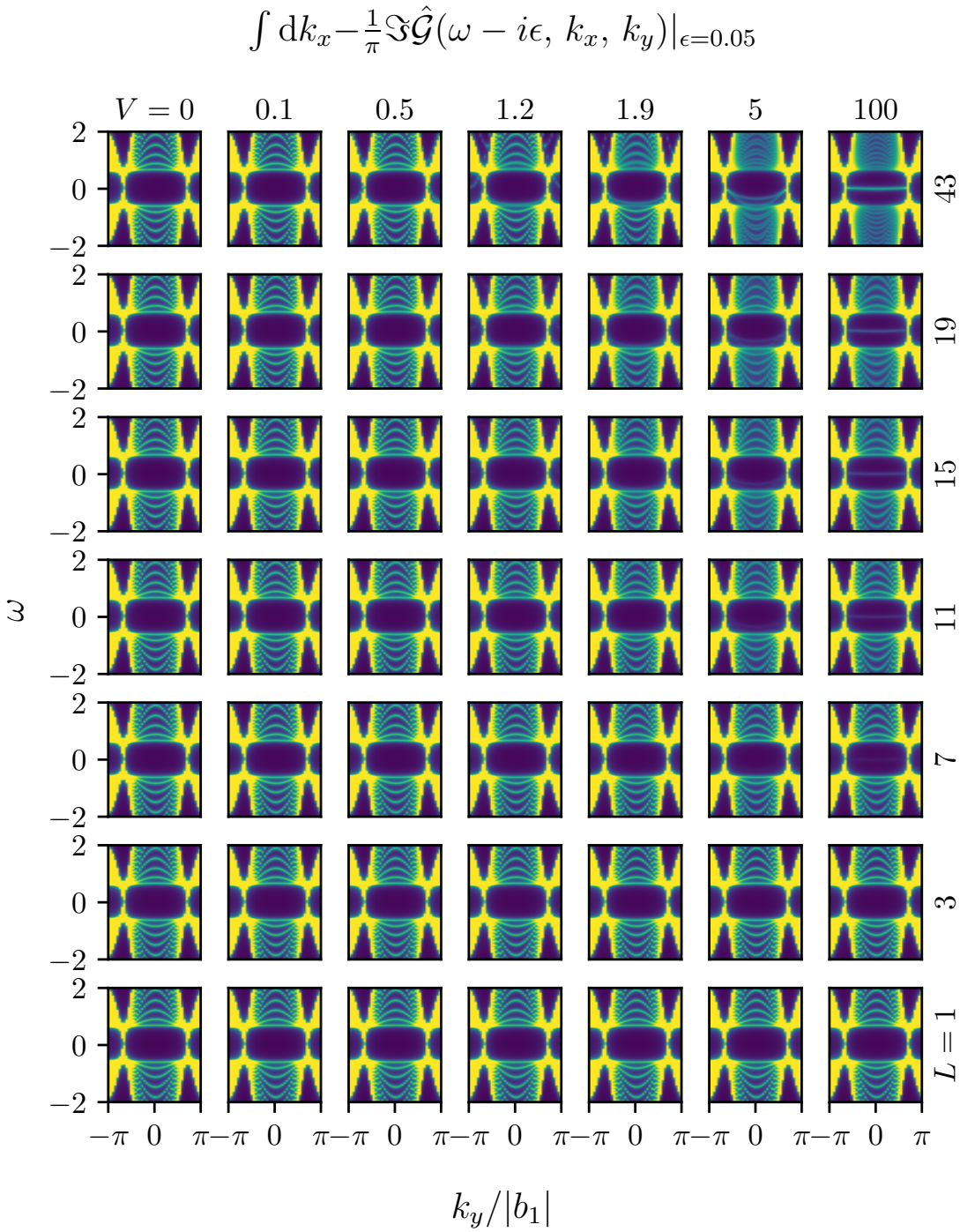

Figure 18: Phase diagram of an impurity wall, perpendicular to the SSH dimers, of length L and coupling strength V, and with closed boundary conditions. The spectrum of the simple Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites. The model consists of an 43×43 lattice, with chemical potential $\mu = 0.0$, interdimer $w = 1.2$, intradimer $v = 0.6$ and interchain $t_d = 0.9$ hoppings. Compare with the model with open boundary conditions Fig 19.

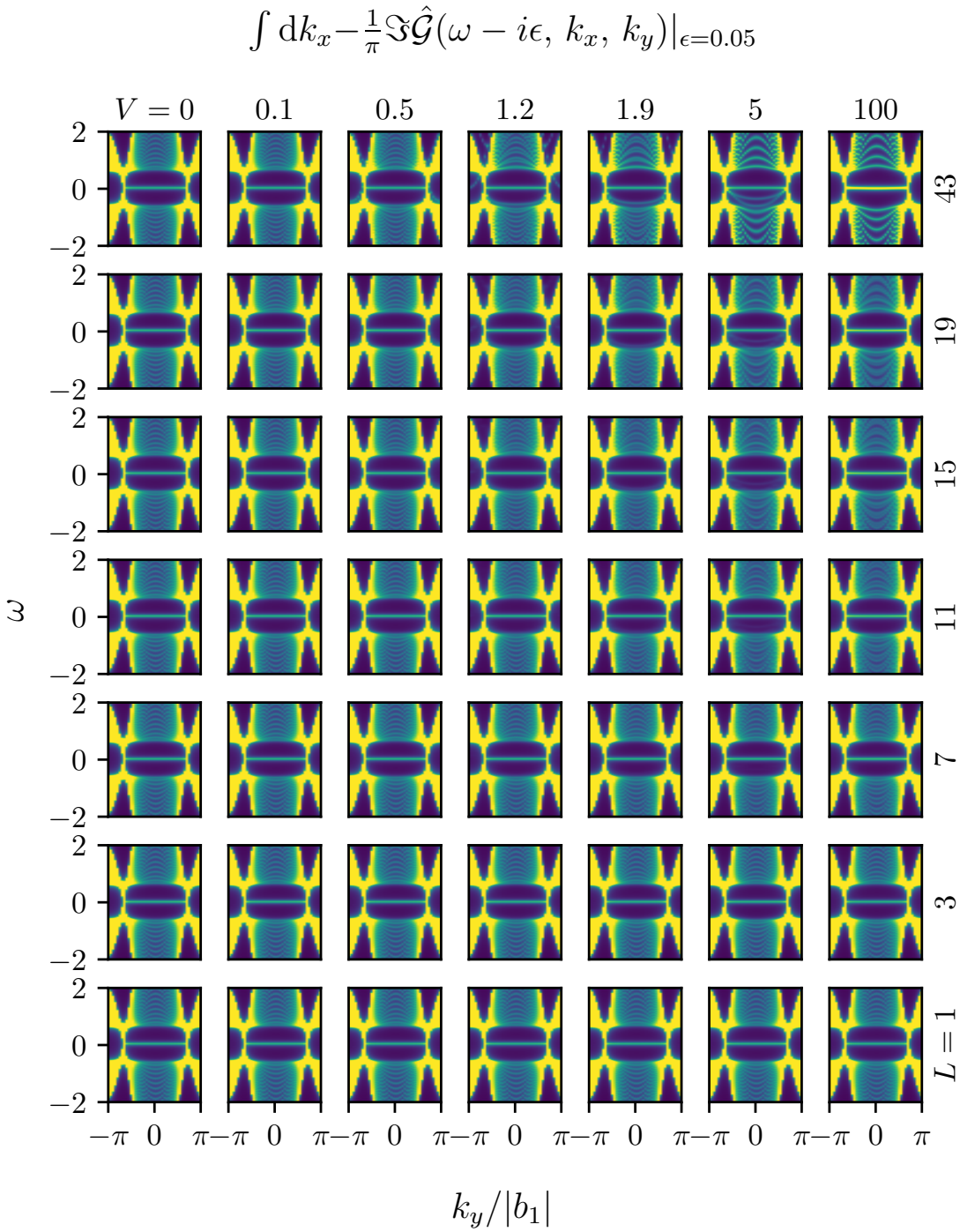

Figure 19: The model and quantities depicted as in Fig 18, but with the boundary conditions open. We observe the topological mode due to the impurity is independent of the topological modees assoicated with the boundary conditions.

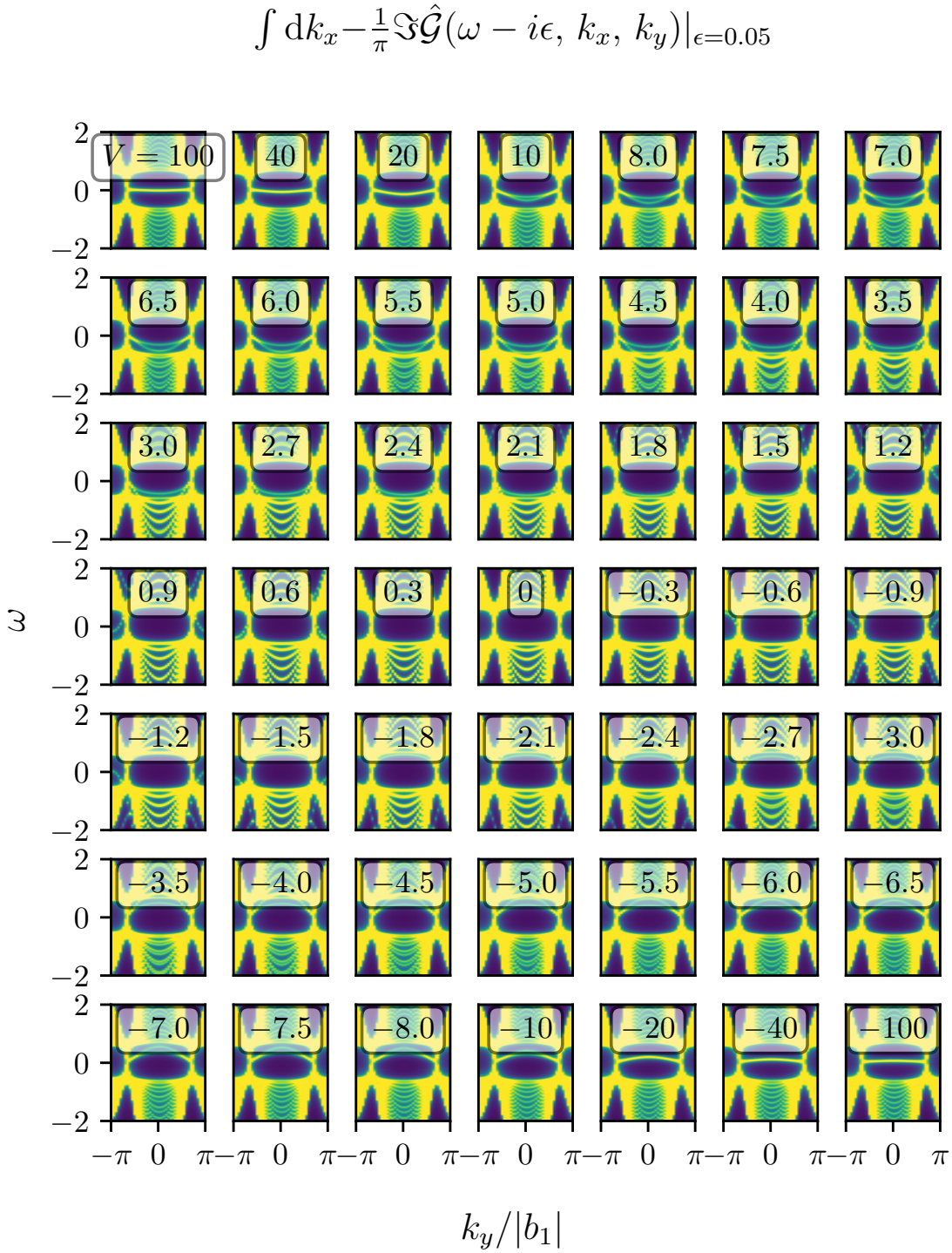

Figure 20: Impurity wall. The spectrum of the Weyl-SSH model in the normal state, in k-space, integrated over the direction along the SSH chains and summed over the atomic sites, 43×43 cells, with closed boundary conditions. The chemical potential is $\mu = 0.0$, interdimer hopping $w = 1.2$, intradimer $v = 0.6$ and interchain $t_d = 0.9$, corresponding to the topological phase of the SSH model. The potential barrier is attractive with negative $V < 0$ and repulsive with positive $V > 0$. For small $|V| \approx w$, a mode outside the Brillouin Zone leaves the bulk. In particular, zero-energy modes exist at the edge of the Brillouin zone when $|V| = w$. As $|V| > w$ increases, the mode moves into the centre of the Brillouin Zone, falling or rising with the sign of V, tending to the topological spectrum of the open-

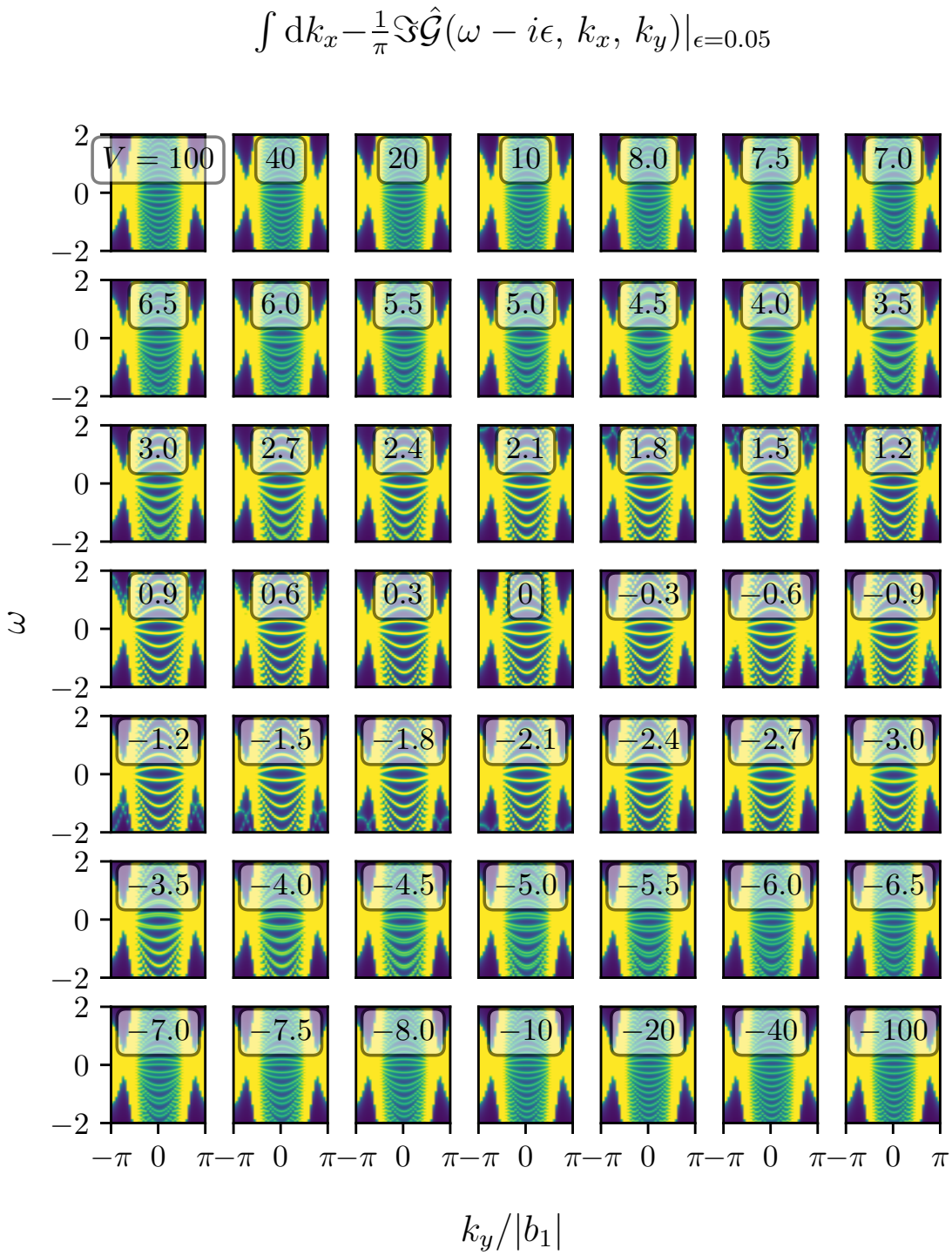

Figure 21: Model and quantities depicted in Figure 20, except with the interdimer hopping $v = w = 1.2$, the topological transition point of the SSH model.

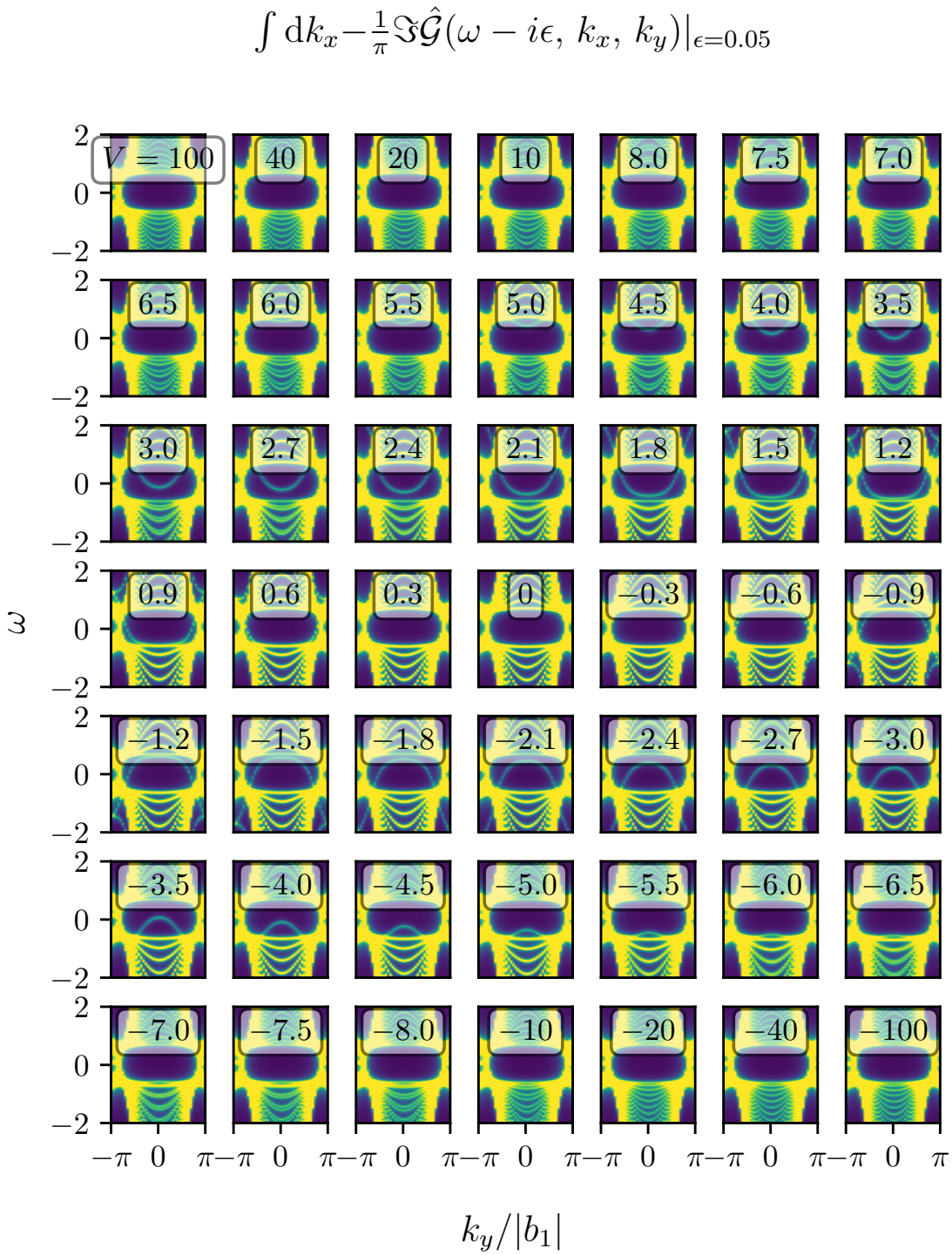

Figure 22: Model and quantities depicted in Figure 20, except with the interdimer hopping $v = 1.8 > w$, the trivial phase of the SSH model.